

Erdős Problem #365

EP-365

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Do all pairs of consecutive powerful numbers n and $n + 1$ come from solutions to Pell equations? In other words, must either n or $n + 1$ be a square?

Is the number of such $n \leq x$ bounded by $(\log x)^{O(1)}$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsignedMath>